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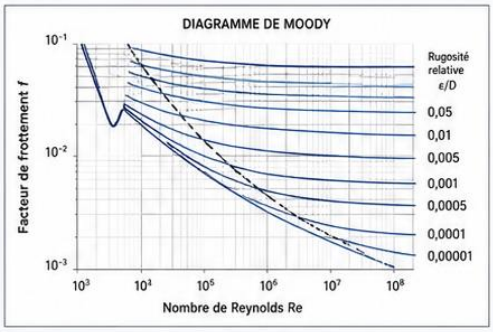
ABAQUES ET TABLEAUX

OUTILS DE RÉFÉRENCE EN MÉCANIQUE DES FLUIDES ET AÉRODYNAMIQUE

LES PERTES DE CHARGE
dans les conduites

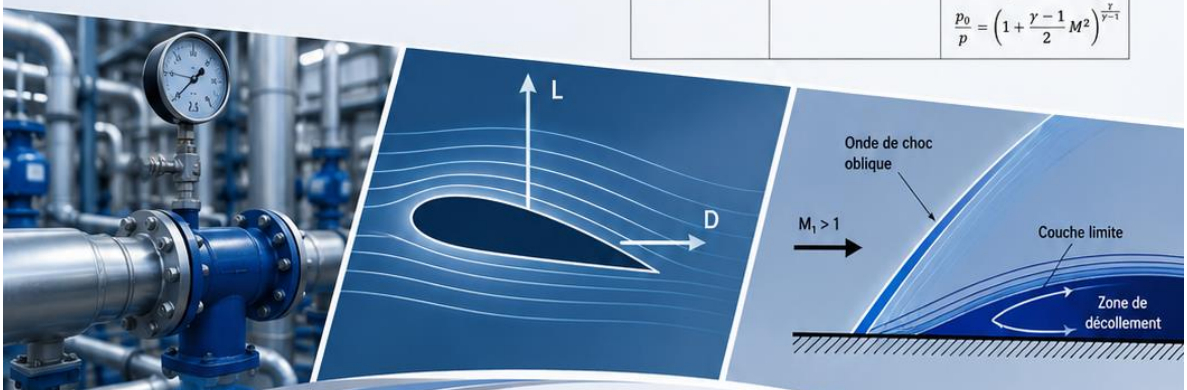
LES FORCES DE PORTANCE ET DE TRAÎNÉE

LES ÉCOULEMENTS COMPRESSIBLES ADIABATIQUES



RÉSUMÉ DES FORMULES UTILISÉES

Sujet	Grandeur	Formule principale
Pertes de charge dans les conduites	Perte de charge linéaire	$h_f = f\frac{L}{D}\frac{V^2}{2g}$
Forces de portance et de traînée	Portance, Traînée	$L = C_L \frac{1}{2}\rho V^2 S$ $D = C_D \frac{1}{2}\rho V^2 S$
Écoulements compressibles adiabatiques	Relations isentropiques	$\frac{T_0}{T} = \left(1+\frac{\gamma-1}{2}M^2\right)$ $\frac{p_0}{p} = \left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}}$



DES DONNÉES ESSENTIELLES POUR L'INGÉNIEUR

Module	Filière	Semestre
Dynamique des Fluides	Génie Mec Prod Th	3ème

Pr. Abdellah OUSEGUI

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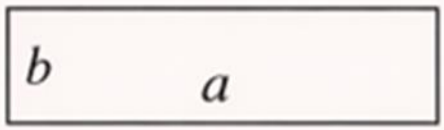
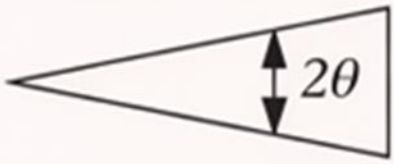
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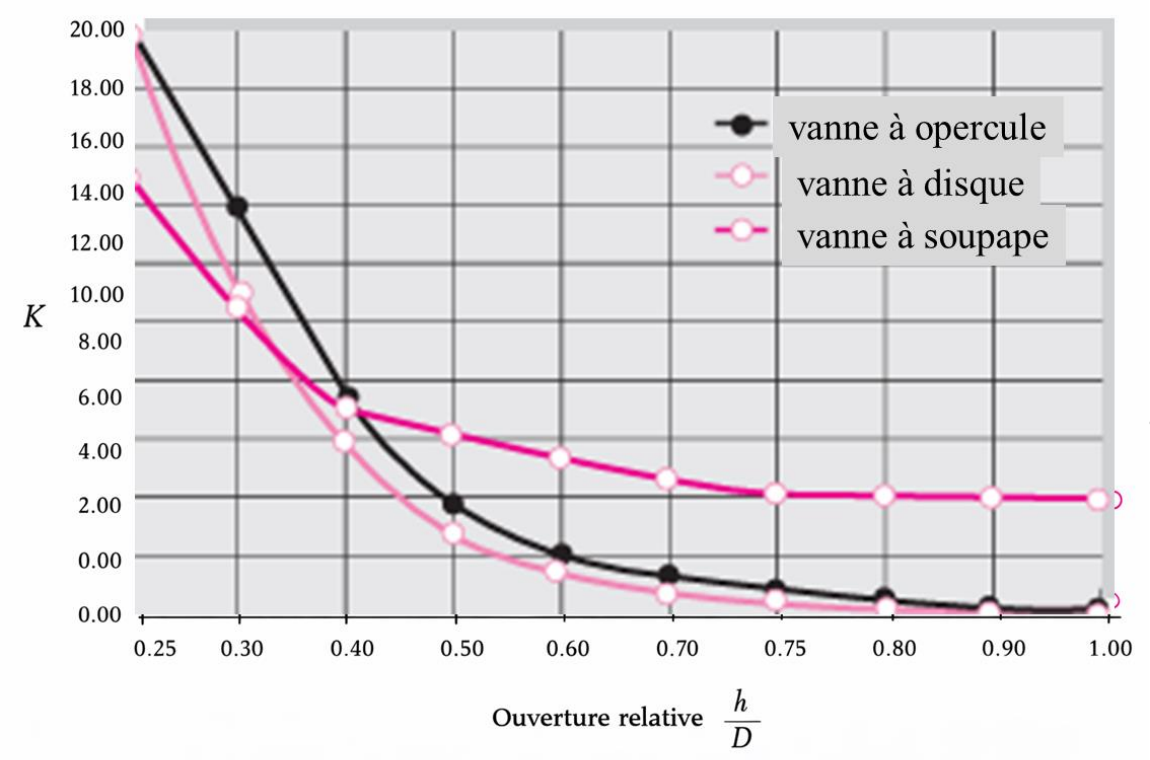
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Annexe 1 : Ecoulement visqueux dans les conduites

Rectangulaire		Triangle isocèle	
			
b/a	$f\,Re_{D_h}$	θ (degrés)	$f\,Re_{D_h}$
0.0	96.00	0	48.0
0.05	89.91	10	51.6
0.1	84.68	20	52.9
0.125	82.34	30	53.3
0.167	78.81	40	52.9
0.25	72.93	50	52.0
0.5	62.19	60	51.1
0.75	57.89	70	49.5
0.75	57.89	70	49.5
1.0	56.91	90	48.0

Facteur de frottement pour des conduites non circulaires

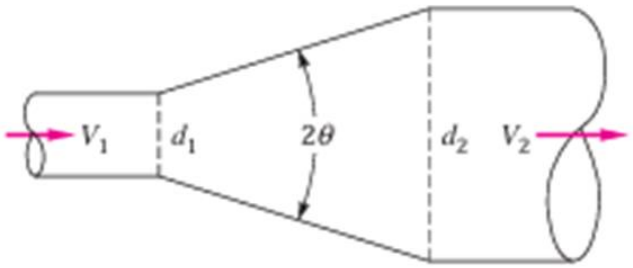
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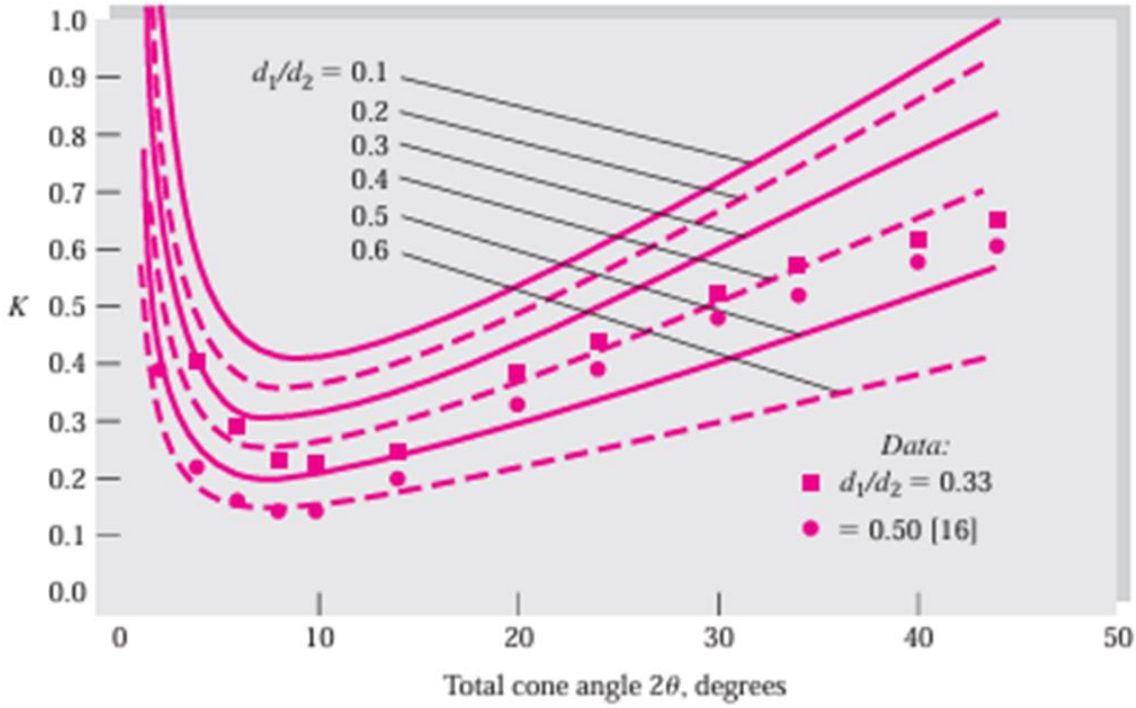


Coefficient de frottement $K = \frac{h_m}{V^2/(2g)}$ pour différents types de vannes

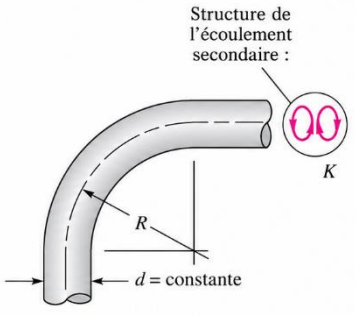
	Diamètre nominal, mm									
	À visser					À brides				
	$\frac{1}{2}$	15	25	40	50	25	40	100	8	500
Vannes (ouvertes):										
Globe	14	8.2	6.9	5.7	13	8.5	6.0	6.0	5.5	5.5
Clapet	0.30	0.24	0.16	0.11	0.80	0.35	0.16	0.16	0.07	0.07
Anti-retour à battant	5.1	2.9	2.1	2.0	3.4	2.5	2.0	2.0	2.0	2.0
Angle	9.0	4.7	2.0	2.0	4.5	2.4	2.4	2.0	2.0	2.0
Coudes:										
45° ordinaire	0.39	0.32	0.30	0.29	0.21	0.20	0.19	0.19	0.14	0.14
45° grand rayon	0.20	0.15	0.95	0.64	0.50	0.35	0.20	0.19	0.19	0.14
90° ordinaire	2.0	1.0	0.94	0.75	1.00	0.50	0.36	0.35	0.35	0.25
90° grand rayon	1.0	0.72	0.41	0.23	0.40	0.21	0.21	0.20	0.20	0.19
180° ordinaire	2.0	1.5	1.0	0.84	1.0	0.40	0.35	0.30	0.30	0.14
180° grand rayon	1.0	0.90	0.50	0.41	0.51	0.20	0.19	0.19	0.19	0.14
Tés:										
Passage direct	0.90	0.90	0.90	0.90	0.24	0.19	0.14	0.07		
Passage latéral	2.4	1.8	1.4	1.1	1.0	0.80	0.64	0.41		
Tés:										
Passage direct	0.90	0.90	0.90	0.90	0.24	0.19	0.14	0.07		
Passage latéral	2.4	1.8	1.4	1.1	1.0	0.80	0.64	0.41		

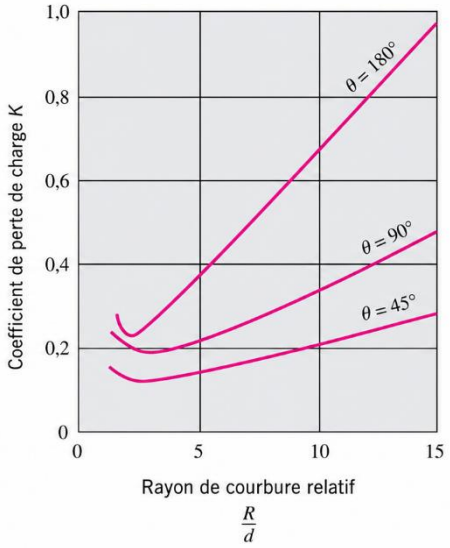
Coefficients de perte de charge $K = \frac{h_m}{V^2/(2g)}$ pour les vannes ouvertes, coudes et tés.





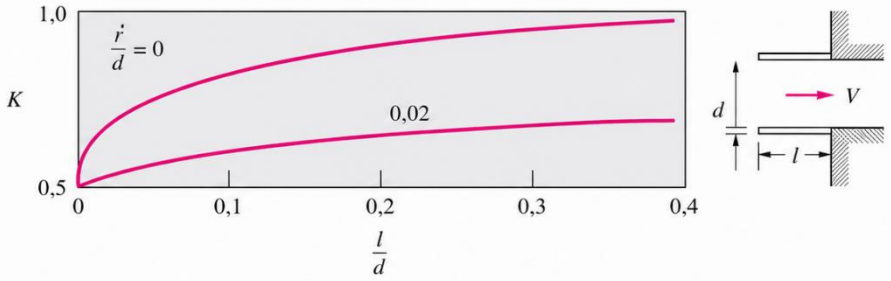
Pertes de charge dans une zone d'expansion conique progressive



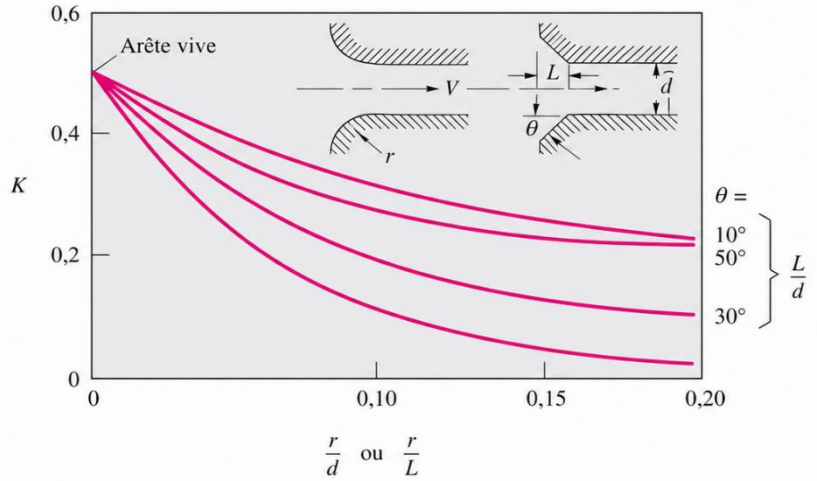


Coefficients de perte de charge pour des coudes de 45°, 90° et 180° à paroi lisse, pour $Re_d = 200\,000$, selon Ito [49].

(a) Entrées réentrantes (coefficient de perte K en fonction du rapport l/d)



(b) Entrées arrondies ou chanfreinées (coefficiess de perte K en fonction du rapport r/d ou r/L)
Les pertes de sortie sont $K \approx 1{,}0$ pour toutes les formes de sortie (réentrante, chanfreinée ou arrondie).
(D'après Réf. 37.)



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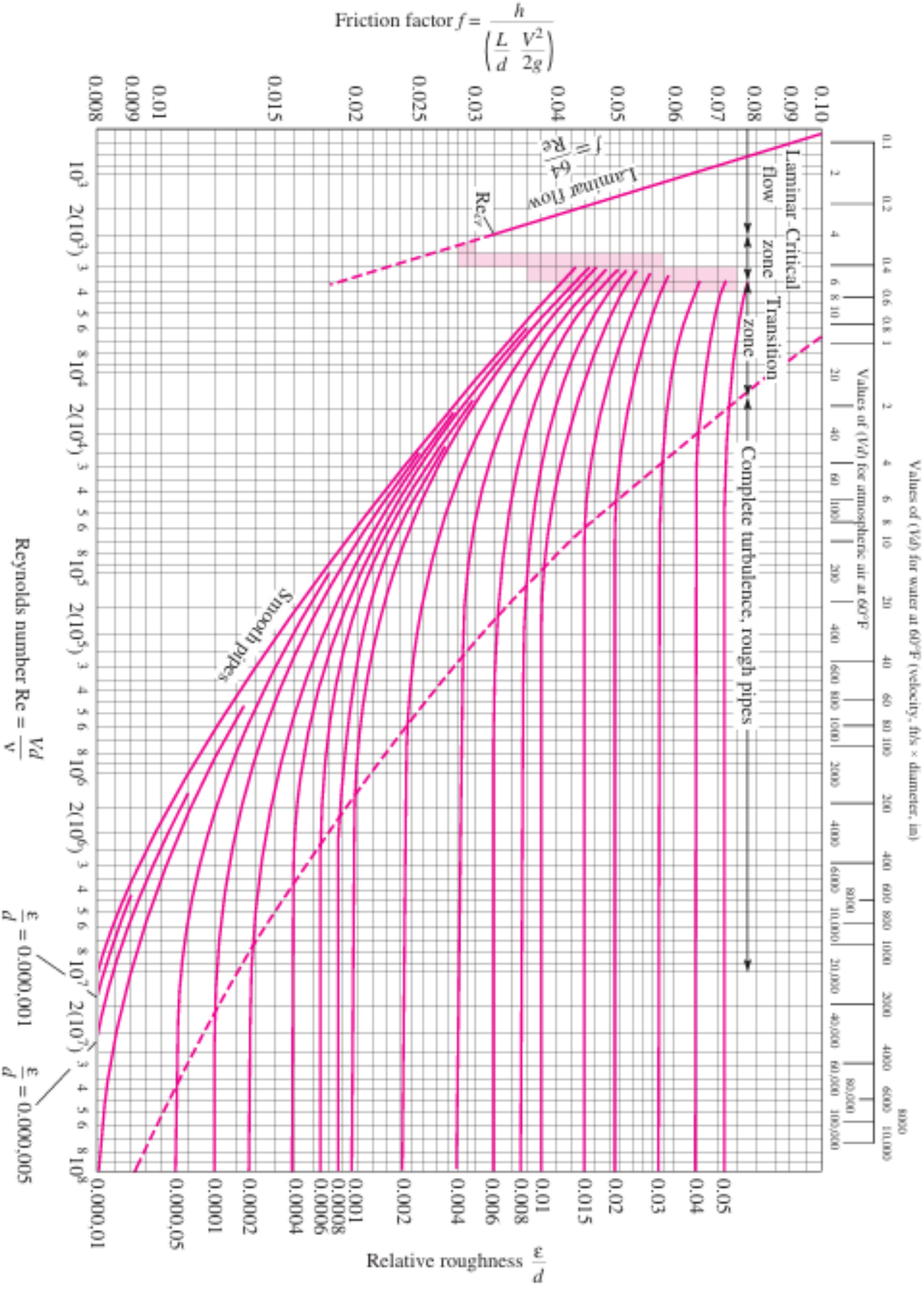


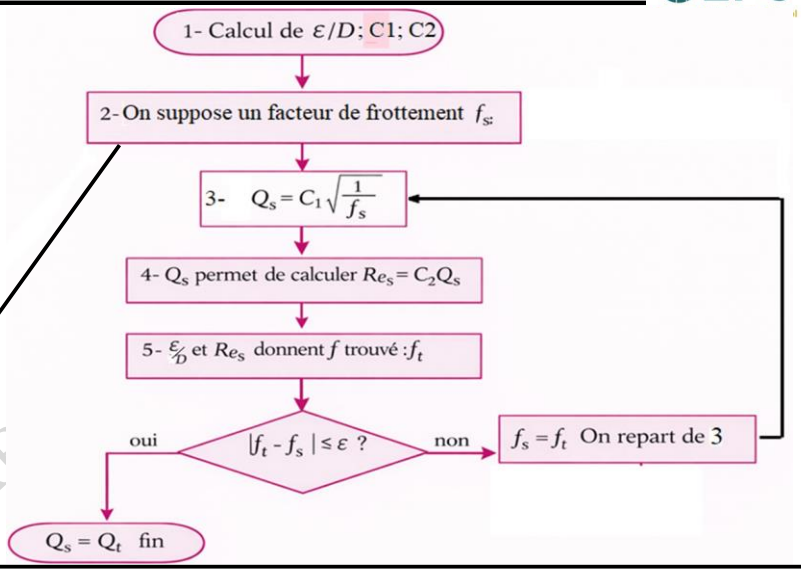
Diagramme de Moody

Problème type 2

$$C1 = \frac{\pi D^2}{4}\sqrt{\frac{2gh_f D}{L}}$$

$$C2 = \frac{4}{\pi \nu D}$$

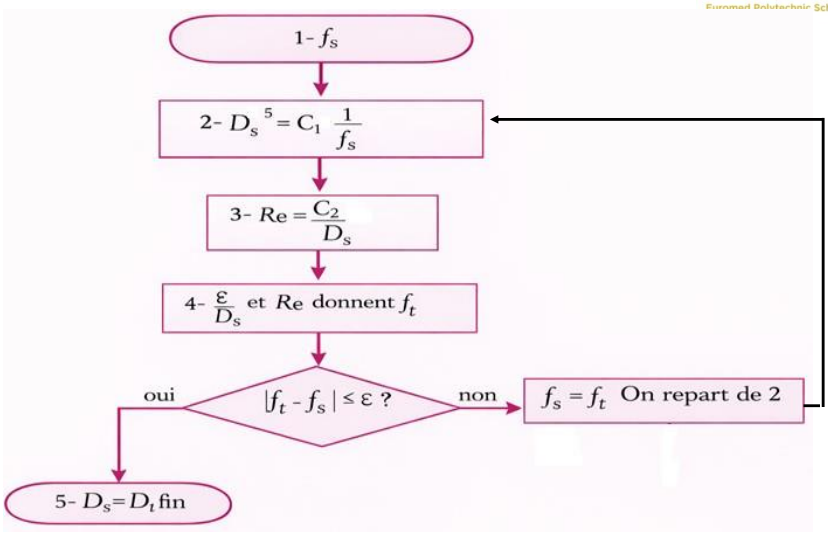
$$\frac{1}{\sqrt{f}} = -2{,}0 \ln\left(\frac{\epsilon/D}{3{,}71}\right)$$



Problème type 3

$$C1 = \frac{8LQ^2}{\pi^2 g h_f}$$

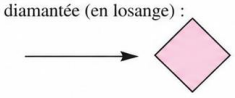
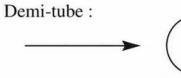
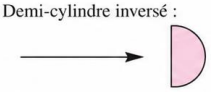
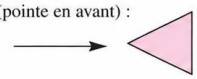
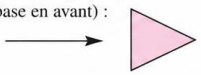
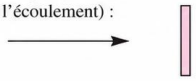
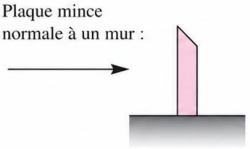
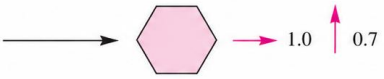
$$C2 = \frac{4Q}{\pi \nu}$$



Résolution des problèmes de détermination du débit (Pb2) et de diamètre (Pb3)

Annexe 2 : Ecoulement externe autour d'un obstacle

Coefficients de traînée C_D basés sur l'aire frontale

Forme	C_D basé sur l'aire frontale
Cylindre à section carrée : 	2.1
Cylindre à section diamantée (en losange) : 	1.6
Demi-tube : 	1.2
Tube circulaire (cylindre creux) : 	2.3
Demi-cylindre : 	1.2
Demi-cylindre inversé : 	1.7
Triangle équilatéral (pointe en avant) : 	1.6
Triangle équilatéral (base en avant) : 	2.0
Plaque (normale à l'écoulement) : 	2.0
Plaque mince normale à un mur : 	1.4
Hexagone : 	1.0 / 0.7

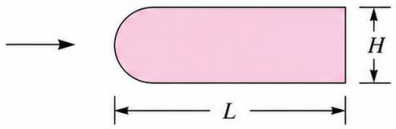
Remarque : Le coefficient de traînée C_D est défini par : $C_D = \dfrac{F_D}{\frac{1}{2}\rho V^2 A_f}$ où F_D est la force de traînée, ρ la masse volumique du fluide, V la vitesse de l'écoulement et A_f l'aire frontale de la forme considérée.

→ Écoulement horizontal

↑ Écoulement vertical

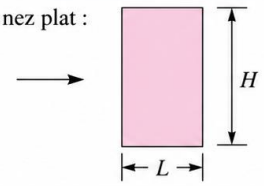
Forme — Coefficient de traînée C_D basé sur la surface frontale

Section à nez arrondi :



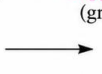
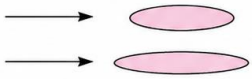
L/H :	0.5	1.0	2.0	4.0	6.0
C_D :	1.16	0.90	0.70	0.68	0.64

Section à nez plat :



L/H :	0.1	0.4	0.7	1.2	2.0	2.5	3.0	6.0
C_D :	1.9	2.3	2.7	2.1	1.8	1.4	1.3	0.9

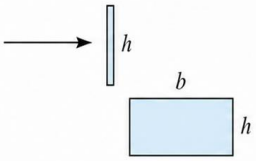
Cylindre elliptique :

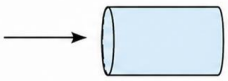
Rapport des axes (grand : petit)	Écoulement laminaire	Écoulement turbulent
1 : 1 	1.2	0.3
2 : 1 	0.6	0.2
4 : 1 	0.35	0.15
8 : 1 	0.25	0.10

Remarque : Le coefficient de traînée C_D est défini par : $C_D = \dfrac{F_D}{\frac{1}{2}\rho V^2 A_f}$ où F_D est la force de traînée (N), ρ la masse volumique du fluide (kg/m^3), V la vitesse d'écoulement (m/s) et A_f la surface frontale (m^2).

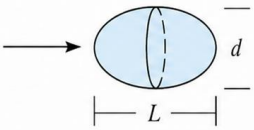
→ Direction de l'écoulement

Traînée pour des corps 2D à $Re \geq 10^4$

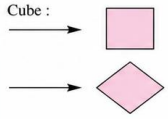
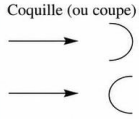
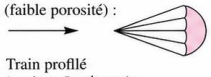
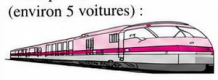
Corps	Rapport	C_D basé sur la surface frontale
Plaque rectangulaire :		
	b/h 1	1.18
	5	1.2
	10	1.3
	20	1.5
	∞	2.0

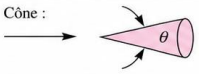
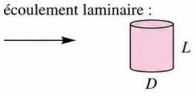
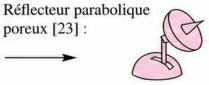
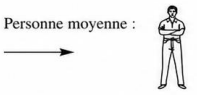
Corps	Rapport	C_D basé sur la surface frontale
Cylindre à face plane :		
	L/d 0.5	1.15
	1	0.90
	2	0.85
	4	0.87
	8	0.99

Ellipsoïde :

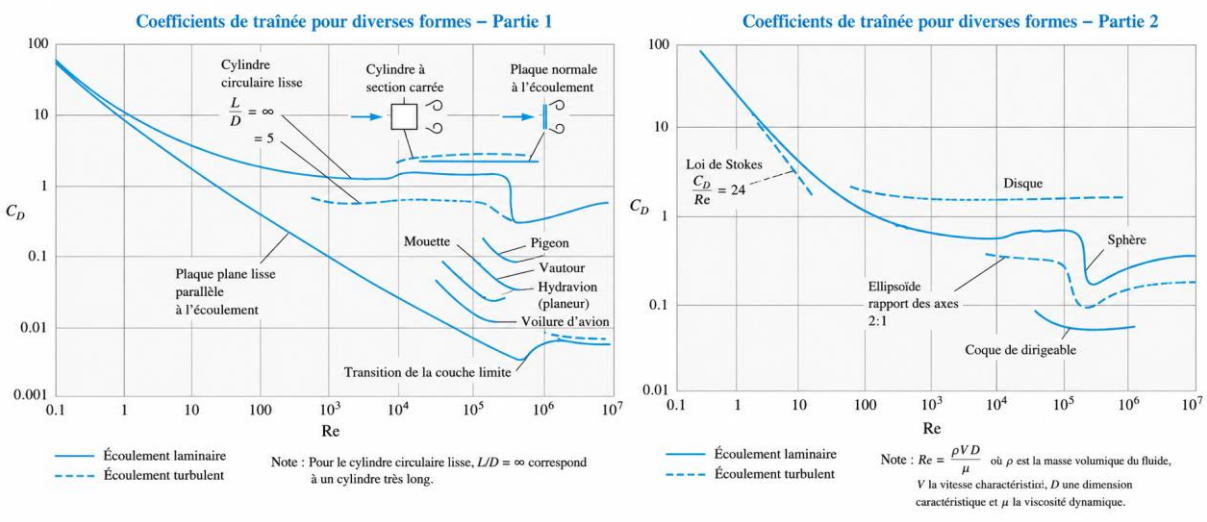


L/d	Écoulement laminaire	Écoulement turbulent
0.75	0.5	0.2
1	0.47	0.2
2	0.27	0.13
4	0.25	0.1
8	0.2	0.08

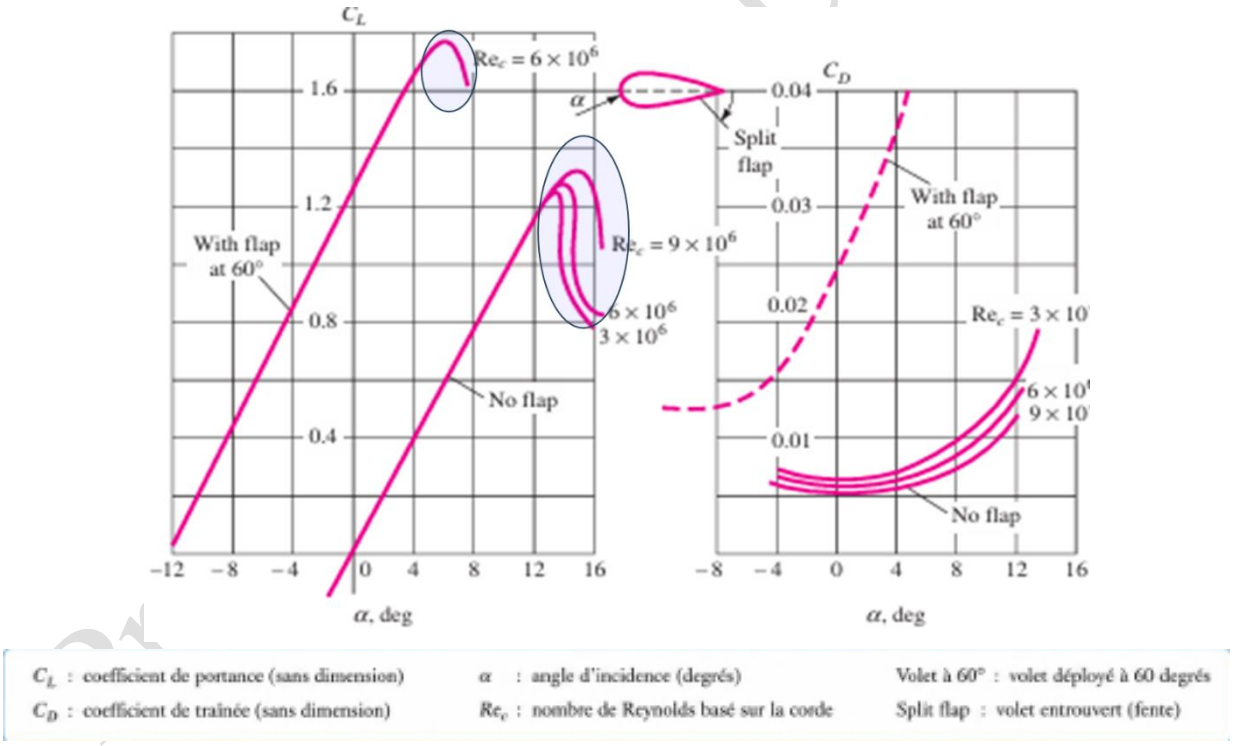
Corps	Coefficient de traînée C_D basé sur la surface frontale
Cube : 	1.07
	0.81
Coquille (ou coupe) : 	1.4
	0.4
Disque : 	1.17
Parachute (faible porosité) : 	1.2
Train profilé (environ 5 voitures) : 	C_DA = 8.5 m²
Vélo : 	Position droite : C_DA = 0.51 m² ; Position de course : C_DA = 0.30 m²

Corps	Coefficient de traînée C_D basé sur la surface frontale
Cône : 	θ : 10° 20° 30° 40° 60° 75° 90° ; C_D : 0.30 0.40 0.55 0.65 0.80 1.05 1.15
Cylindre court, écoulement laminaire : 	L/D : 1 2 3 5 10 20 40 ∞ ; C_D : 0.64 0.68 0.72 0.74 0.82 0.91 0.98 1.20
Réflecteur parabolique poreux [23] : 	Porosité : 0 0.1 0.2 0.3 0.4 0.5 ; ← C_D : 1.42 1.33 1.20 1.05 0.95 0.82 ; → C_D : 0.95 0.92 0.90 0.86 0.83 0.80
Personne moyenne : 	→ C_DA = 9 ft² ↑ C_DA = 1.2 ft²
Pins et épicéas [24] : 	U (m/s) : 10 20 30 40 ; C_D : 1.2 ± 0.2 1.0 ± 0.2 0.7 ± 0.2 0.5 ± 0.2
Camion semi-remorque : 	Sans déflecteur : 0.96 ; avec déflecteur : 0.76

Traînée pour des corps 3D à $Re \geq 10^4$



Évolution du coefficient de traînée C_D en fonction du nombre de Reynolds pour différentes géométries de corps



Variation des coefficients aérodynamiques C_L et C_D en fonction de l'angle d'attaque pour un profil avec et sans volet

Annexe 3 : Ecoulement compressible isentropique

Pr. Abdellah OUSEGUI

TABLE I : Ecoulement isentropique d'un gaz parfait (γ=1.4).

M	p/p_0	ρ/ρ_0	T/T_0	A/A^*
0.00	1.0000	1.0000	1.0000	∞
0.02	0.9997	0.9998	0.9999	28.9421
0.04	0.9989	0.9992	0.9997	14.4815
0.06	0.9975	0.9982	0.9993	9.6659
0.08	0.9955	0.9968	0.9987	7.2616
0.10	0.9930	0.9950	0.9980	5.8218
0.12	0.9900	0.9928	0.9971	4.8643
0.14	0.9864	0.9903	0.9961	4.1824
0.16	0.9823	0.9873	0.9949	3.6727
0.18	0.9776	0.9840	0.9936	3.2779
0.20	0.9725	0.9803	0.9921	2.9635
0.22	0.9668	0.9762	0.9904	2.7076
0.24	0.9607	0.9718	0.9886	2.4956
0.26	0.9541	0.9670	0.9867	2.3173
0.28	0.9470	0.9619	0.9846	2.1656
0.30	0.9395	0.9564	0.9823	2.0351
0.32	0.9315	0.9506	0.9799	1.9219
0.34	0.9231	0.9445	0.9774	1.8229
0.36	0.9143	0.9380	0.9747	1.7358
0.38	0.9052	0.9313	0.9719	1.6587
0.40	0.8956	0.9243	0.9690	1.5901
0.42	0.8857	0.9170	0.9659	1.5289
0.44	0.8755	0.9094	0.9627	1.4740
0.46	0.8650	0.9016	0.9594	1.4246
0.48	0.8541	0.8935	0.9559	1.3801
0.50	0.8430	0.8852	0.9524	1.3398
0.52	0.8317	0.8766	0.9487	1.3034
0.54	0.8201	0.8679	0.9449	1.2703
0.56	0.8082	0.8589	0.9410	1.2403
0.58	0.7962	0.8498	0.9370	1.2130
0.60	0.7840	0.8405	0.9328	1.1882
0.62	0.7716	0.8310	0.9286	1.1656
0.64	0.7591	0.8213	0.9243	1.1451
0.66	0.7465	0.8115	0.9199	1.1265
0.68	0.7338	0.8016	0.9153	1.1097
0.70	0.7209	0.7916	0.9107	1.0944
0.72	0.7080	0.7814	0.9061	1.0806
0.74	0.6951	0.7712	0.9013	1.0681
0.76	0.6821	0.7609	0.8964	1.0570
0.78	0.6691	0.7505	0.8915	1.0471
0.80	0.6560	0.7400	0.8865	1.0382
0.82	0.6430	0.7295	0.8815	1.0305
0.84	0.6300	0.7189	0.8763	1.0237
0.86	0.6170	0.7083	0.8711	1.0179
0.88	0.6041	0.6977	0.8659	1.0129
0.90	0.5913	0.6870	0.8606	1.0089
0.92	0.5785	0.6764	0.8552	1.0056
0.94	0.5658	0.6658	0.8498	1.0031
0.96	0.5532	0.6551	0.8444	1.0014
0.98	0.5407	0.6445	0.8389	1.0003
1.00	0.5283	0.6339	0.8333	1.0000
1.02	0.5160	0.6234	0.8278	1.0003
1.04	0.5039	0.6129	0.8222	1.0013
1.06	0.4919	0.6024	0.8165	1.0029
1.08	0.4800	0.5920	0.8108	1.0051
1.10	0.4684	0.5817	0.8052	1.0079
1.12	0.4568	0.5714	0.7994	1.0113
1.14	0.4455	0.5612	0.7937	1.0153
1.16	0.4343	0.5511	0.7879	1.0198
1.18	0.4232	0.5411	0.7822	1.0248
1.20	0.4124	0.5311	0.7764	1.0304
1.22	0.4017	0.5213	0.7706	1.0366
1.24	0.3912	0.5115	0.7648	1.0432
1.26	0.3809	0.5019	0.7590	1.0504
1.28	0.3708	0.4923	0.7532	1.0581
1.30	0.3609	0.4829	0.7474	1.0663
1.32	0.3512	0.4736	0.7416	1.0750
1.34	0.3417	0.4644	0.7358	1.0842
1.36	0.3323	0.4553	0.7300	1.0940
1.38	0.3232	0.4463	0.7242	1.1042
1.40	0.3142	0.4374	0.7184	1.1149
1.42	0.3055	0.4287	0.7126	1.1262
1.44	0.2969	0.4201	0.7069	1.1379
1.46	0.2886	0.4116	0.7011	1.1501
1.48	0.2804	0.4032	0.6954	1.1629
1.50	0.2724	0.3950	0.6897	1.1762
1.52	0.2646	0.3869	0.6840	1.1899
1.54	0.2570	0.3789	0.6783	1.2042
1.56	0.2496	0.3710	0.6726	1.2190
1.58	0.2423	0.3633	0.6670	1.2344
1.60	0.2353	0.3557	0.6614	1.2502
1.62	0.2284	0.3483	0.6558	1.2666
1.64	0.2217	0.3409	0.6502	1.2836
1.66	0.2152	0.3337	0.6447	1.3010
1.68	0.2088	0.3266	0.6392	1.3190
1.70	0.2026	0.3197	0.6337	1.3376
1.72	0.1966	0.3129	0.6283	1.3567
1.74	0.1907	0.3062	0.6229	1.3764
1.76	0.1850	0.2996	0.6175	1.3967
1.78	0.1794	0.2931	0.6121	1.4175
1.80	0.1740	0.2868	0.6068	1.4390
1.82	0.1688	0.2806	0.6015	1.4610

TABLE I (Suite): Ecoulement isentropique d'un gaz parfait ($\gamma = 1.4$)

M	p / p_0	ρ / ρ_0	T / T_0	A / A^*
1.84	0.1637	0.2745	0.5963	1.4836
1.86	0.1587	0.2686	0.5910	1.5069
1.88	0.1539	0.2627	0.5859	1.5308
1.90	0.1492	0.2570	0.5807	1.5553
1.92	0.1447	0.2514	0.5756	1.5804
1.94	0.1403	0.2459	0.5705	1.6062
1.96	0.1360	0.2405	0.5655	1.6326
1.98	0.1318	0.2352	0.5605	1.6597
2.00	0.1278	0.2300	0.5556	1.6875
2.02	0.1239	0.2250	0.5506	1.7160
2.04	0.1201	0.2200	0.5458	1.7451
2.06	0.1164	0.2152	0.5409	1.7750
2.08	0.1128	0.2104	0.5361	1.8056
2.10	0.1094	0.2058	0.5313	1.8369
2.12	0.1060	0.2013	0.5266	1.8690
2.14	0.1027	0.1968	0.5219	1.9018
2.16	0.0996	0.1925	0.5173	1.9354
2.18	0.0965	0.1882	0.5127	1.9698
2.20	0.0935	0.1841	0.5081	2.0050
2.22	0.0906	0.1800	0.5036	2.0409
2.24	0.0878	0.1760	0.4991	2.0777
2.26	0.0851	0.1721	0.4947	2.1153
2.28	0.0825	0.1683	0.4903	2.1538
2.30	0.0800	0.1646	0.4859	2.1931
2.32	0.0775	0.1609	0.4816	2.2333
2.34	0.0751	0.1574	0.4773	2.2744
2.36	0.0728	0.1539	0.4731	2.3164
2.38	0.0706	0.1505	0.4688	2.3593
2.40	0.0684	0.1472	0.4647	2.4031
2.42	0.0663	0.1439	0.4606	2.4479
2.44	0.0643	0.1408	0.4565	2.4936
2.46	0.0623	0.1377	0.4524	2.5403
2.48	0.0604	0.1346	0.4484	2.5880
2.50	0.0585	0.1317	0.4444	2.6367
2.52	0.0567	0.1288	0.4405	2.6864
2.54	0.0550	0.1260	0.4366	2.7372
2.56	0.0533	0.1232	0.4328	2.7891
2.58	0.0517	0.1205	0.4289	2.8420
2.60	0.0501	0.1179	0.4252	2.8960
2.62	0.0486	0.1153	0.4214	2.9511
2.64	0.0471	0.1128	0.4177	3.0073
2.66	0.0457	0.1103	0.4141	3.0647
2.68	0.0443	0.1079	0.4104	3.1233
2.70	0.0430	0.1056	0.4068	3.1830
2.72	0.0417	0.1033	0.4033	3.2439
2.74	0.0404	0.1010	0.3998	3.3061
2.76	0.0392	0.0989	0.3963	3.3695
2.78	0.0380	0.0967	0.3928	3.4342
2.80	0.0368	0.0946	0.3894	3.5001
2.82	0.0357	0.0926	0.3860	3.5674
2.84	0.0347	0.0906	0.3827	3.6359
2.86	0.0336	0.0886	0.3794	3.7058
2.88	0.0326	0.0867	0.3761	3.7771
2.90	0.0317	0.0849	0.3729	3.8498
2.92	0.0307	0.0831	0.3696	3.9238
2.94	0.0298	0.0813	0.3665	3.9993
2.96	0.0289	0.0796	0.3633	4.0762
2.98	0.0281	0.0779	0.3602	4.1547
3.00	0.0272	0.0762	0.3571	4.2346
3.02	0.0264	0.0746	0.3541	4.3160
3.04	0.0256	0.0730	0.3511	4.3989
3.06	0.0249	0.0715	0.3481	4.4835
3.08	0.0242	0.0700	0.3452	4.5696
3.10	0.0234	0.0685	0.3422	4.6573
3.12	0.0228	0.0671	0.3393	4.7467
3.14	0.0221	0.0657	0.3365	4.8377
3.16	0.0215	0.0643	0.3337	4.9304
3.18	0.0208	0.0630	0.3309	5.0248
3.20	0.0202	0.0617	0.3281	5.1209
3.22	0.0196	0.0604	0.3253	5.2189
3.24	0.0191	0.0591	0.3226	5.3186
3.26	0.0185	0.0579	0.3199	5.4201
3.28	0.0180	0.0567	0.3173	5.5234
3.30	0.0175	0.0555	0.3147	5.6286
3.32	0.0170	0.0544	0.3121	5.7357
3.34	0.0165	0.0533	0.3095	5.8448
3.36	0.0160	0.0522	0.3069	5.9558
3.38	0.0156	0.0511	0.3044	6.0687
3.40	0.0151	0.0501	0.3019	6.1837
3.42	0.0147	0.0491	0.2995	6.3007
3.44	0.0143	0.0481	0.2970	6.4197
3.46	0.0139	0.0471	0.2946	6.5409
3.48	0.0135	0.0462	0.2922	6.6642
3.50	0.0131	0.0452	0.2899	6.7896
3.52	0.0127	0.0443	0.2875	6.9172
3.54	0.0124	0.0434	0.2852	7.0470
3.56	0.0120	0.0426	0.2829	7.1791
3.58	0.0117	0.0417	0.2806	7.3134
3.60	0.0114	0.0409	0.2784	7.4501
3.62	0.0111	0.0401	0.2762	7.5891
3.64	0.0108	0.0393	0.2740	7.7304
3.66	0.0105	0.0385	0.2718	7.8742
3.68	0.0102	0.0378	0.2697	8.0204
3.70	0.0099	0.0370	0.2675	8.1690
3.72	0.0096	0.0363	0.2654	8.3202
3.74	0.0094	0.0356	0.2633	8.4739
3.76	0.0091	0.0349	0.2613	8.6302
3.78	0.0089	0.0342	0.2592	8.7891
3.80	0.0086	0.0335	0.2572	8.9506
3.82	0.0084	0.0329	0.2552	9.1147
3.84	0.0082	0.0323	0.2532	9.2816
3.86	0.0080	0.0316	0.2513	9.4513
3.88	0.0077	0.0310	0.2493	9.6237
3.90	0.0075	0.0304	0.2474	9.7989
3.92	0.0073	0.0299	0.2455	9.9770
3.94	0.0071	0.0293	0.2436	10.1580
3.96	0.0069	0.0287	0.2418	10.3419
3.98	0.0068	0.0282	0.2399	10.5288

TABLE I (Suite): Ecoulement isentropique d'un gaz parfait ($\gamma = 1.4$)

M	p / p_0	ρ / ρ_0	T / T_0	A / A^*
4.00	0.0066	0.0277	0.2381	10.7187
4.02	0.0064	0.0271	0.2363	10.9117
4.04	0.0062	0.0266	0.2345	11.1077
4.06	0.0061	0.0261	0.2327	11.3068
4.08	0.0059	0.0256	0.2310	11.5091
4.10	0.0058	0.0252	0.2293	11.7146
4.12	0.0056	0.0247	0.2275	11.9234
4.14	0.0055	0.0242	0.2258	12.1354
4.16	0.0053	0.0238	0.2242	12.3508
4.18	0.0052	0.0234	0.2225	12.5695
4.20	0.0051	0.0229	0.2208	12.7916
4.22	0.0049	0.0225	0.2192	13.0172
4.24	0.0048	0.0221	0.2176	13.2463
4.26	0.0047	0.0217	0.2160	13.4789
4.28	0.0046	0.0213	0.2144	13.7151
4.30	0.0044	0.0209	0.2129	13.9549
4.32	0.0043	0.0205	0.2113	14.1983
4.34	0.0042	0.0202	0.2098	14.4455
4.36	0.0041	0.0198	0.2083	14.6965
4.38	0.0040	0.0194	0.2067	14.9512
4.40	0.0039	0.0191	0.2053	15.2098
4.42	0.0038	0.0187	0.2038	15.4723
4.44	0.0037	0.0184	0.2023	15.7387
4.46	0.0036	0.0181	0.2009	16.0092
4.48	0.0035	0.0178	0.1994	16.2836
4.50	0.0035	0.0174	0.1980	16.5622
4.52	0.0034	0.0171	0.1966	16.8448
4.54	0.0033	0.0168	0.1952	17.1317
4.56	0.0032	0.0165	0.1938	17.4227
4.58	0.0031	0.0163	0.1925	17.7181
4.60	0.0031	0.0160	0.1911	18.0177
4.62	0.0030	0.0157	0.1898	18.3218
4.64	0.0029	0.0154	0.1885	18.6303
4.66	0.0028	0.0152	0.1872	18.9432
4.68	0.0028	0.0149	0.1859	19.2607
4.70	0.0027	0.0146	0.1846	19.5828
4.72	0.0026	0.0144	0.1833	19.9095
4.74	0.0026	0.0141	0.1820	20.2409
4.76	0.0025	0.0139	0.1808	20.5770
4.78	0.0025	0.0137	0.1795	20.9179
4.80	0.0024	0.0134	0.1783	21.2636
4.82	0.0023	0.0132	0.1771	21.6143
4.84	0.0023	0.0130	0.1759	21.9699
4.86	0.0022	0.0128	0.1747	22.3306
4.88	0.0022	0.0125	0.1735	22.6962
4.90	0.0021	0.0123	0.1724	23.0671
4.92	0.0021	0.0121	0.1712	23.4430
4.94	0.0020	0.0119	0.1700	23.8243
4.96	0.0020	0.0117	0.1689	24.2108
4.98	0.0019	0.0115	0.1678	24.6027
5.00	0.0019	0.0113	0.1667	25.0000
∞	0	0	0	∞

TABLE II : Ecoulement à onde de choc normale d'un gaz parfait ($\gamma = 1.4$).

M_1 $M_1 \sin\sigma$	M_2 $M_2 \sin(\sigma - \delta)$	p_2 / p_1	ρ_2 / ρ_1 V_1 / V_2 V_{n1} / V_{n2}	T_2 / T_1	p_{02} / p_{01}	A_2^* / A_1^*	$\frac{\Delta s}{R}$
1.00	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.000
1.02	0.9805	1.0471	1.0334	1.0132	1.0000	1.0000	0.000
1.04	0.9620	1.0952	1.0671	1.0263	0.9999	1.0001	0.000
1.06	0.9444	1.1442	1.1009	1.0393	0.9998	1.0002	0.000
1.08	0.9277	1.1941	1.1349	1.0522	0.9994	1.0006	0.001
1.10	0.9118	1.2450	1.1691	1.0649	0.9989	1.0011	0.001
1.12	0.8966	1.2968	1.2034	1.0776	0.9982	1.0018	0.002
1.14	0.8820	1.3495	1.2378	1.0903	0.9973	1.0027	0.003
1.16	0.8682	1.4032	1.2723	1.1029	0.9961	1.0040	0.004
1.18	0.8549	1.4578	1.3069	1.1154	0.9946	1.0055	0.005
1.20	0.8422	1.5133	1.3416	1.1280	0.9928	1.0073	0.007
1.22	0.8300	1.5698	1.3764	1.1405	0.9907	1.0094	0.009
1.24	0.8183	1.6272	1.4112	1.1531	0.9884	1.0118	0.012
1.26	0.8071	1.6855	1.4460	1.1657	0.9857	1.0145	0.014
1.28	0.7963	1.7448	1.4808	1.1783	0.9827	1.0176	0.017
1.30	0.7860	1.8050	1.5157	1.1909	0.9794	1.0211	0.021
1.32	0.7760	1.8661	1.5505	1.2035	0.9758	1.0249	0.025
1.34	0.7664	1.9282	1.5854	1.2162	0.9718	1.0290	0.029
1.36	0.7572	1.9912	1.6202	1.2290	0.9676	1.0335	0.033
1.38	0.7483	2.0551	1.6549	1.2418	0.9630	1.0384	0.038
1.40	0.7397	2.1200	1.6897	1.2547	0.9582	1.0436	0.043
1.42	0.7314	2.1858	1.7243	1.2676	0.9531	1.0492	0.048
1.44	0.7235	2.2525	1.7589	1.2807	0.9476	1.0552	0.054
1.46	0.7157	2.3202	1.7934	1.2938	0.9420	1.0616	0.060
1.48	0.7083	2.3888	1.8278	1.3069	0.9360	1.0684	0.066
1.50	0.7011	2.4583	1.8621	1.3202	0.9298	1.0755	0.073
1.52	0.6941	2.5288	1.8963	1.3336	0.9233	1.0830	0.080
1.54	0.6874	2.6002	1.9303	1.3470	0.9166	1.0910	0.087
1.56	0.6809	2.6725	1.9643	1.3606	0.9097	1.0993	0.095
1.58	0.6746	2.7458	1.9981	1.3742	0.9026	1.1080	0.103
1.60	0.6684	2.8200	2.0317	1.3880	0.8952	1.1171	0.111
1.62	0.6625	2.8951	2.0653	1.4018	0.8877	1.1266	0.119
1.64	0.6568	2.9712	2.0986	1.4158	0.8799	1.1365	0.128
1.66	0.6512	3.0482	2.1318	1.4299	0.8720	1.1468	0.137
1.68	0.6458	3.1261	2.1649	1.4440	0.8639	1.1575	0.146
1.70	0.6405	3.2050	2.1977	1.4583	0.8557	1.1686	0.156
1.72	0.6355	3.2848	2.2304	1.4727	0.8474	1.1801	0.166
1.74	0.6305	3.3655	2.2629	1.4873	0.8389	1.1921	0.176
1.76	0.6257	3.4472	2.2952	1.5019	0.8302	1.2045	0.186
1.78	0.6210	3.5298	2.3273	1.5167	0.8215	1.2173	0.197
1.80	0.6165	3.6133	2.3592	1.5316	0.8127	1.2305	0.207
1.82	0.6121	3.6978	2.3909	1.5466	0.8038	1.2441	0.218
1.84	0.6078	3.7832	2.4224	1.5617	0.7948	1.2582	0.230
1.86	0.6036	3.8695	2.4537	1.5770	0.7857	1.2728	0.241
1.88	0.5996	3.9568	2.4848	1.5924	0.7765	1.2877	0.253
1.90	0.5956	4.0450	2.5157	1.6079	0.7674	1.3032	0.265
1.92	0.5918	4.1341	2.5463	1.6236	0.7581	1.3191	0.277
1.94	0.5880	4.2242	2.5767	1.6394	0.7488	1.3354	0.289

TABLE II (Suite): Ecoulement à onde de choc normale d'un gaz parfait ($\gamma = 1.4$).

M_1 $M_1 \sin\sigma$	M_2 $M_2 \sin(\sigma - \delta)$	p_2 / p_1	ρ_2 / ρ_1 V_1 / V_2 V_{n1} / V_{n2}	T_2 / T_1	p_{02} / p_{01}	A_2^* / A_1^*	$\frac{\Delta s}{R}$
1.96	0.5844	4.3152	2.6069	1.6553	0.7395	1.3522	0.302
1.98	0.5808	4.4071	2.6369	1.6713	0.7302	1.3695	0.314
2.00	0.5774	4.5000	2.6667	1.6875	0.7209	1.3872	0.327
2.02	0.5740	4.5938	2.6962	1.7038	0.7115	1.4054	0.340
2.04	0.5707	4.6885	2.7255	1.7203	0.7022	1.4241	0.354
2.06	0.5675	4.7842	2.7545	1.7369	0.6928	1.4433	0.367
2.08	0.5643	4.8808	2.7833	1.7536	0.6835	1.4630	0.381
2.10	0.5613	4.9783	2.8119	1.7704	0.6742	1.4832	0.394
2.12	0.5583	5.0768	2.8402	1.7875	0.6649	1.5039	0.408
2.14	0.5554	5.1762	2.8683	1.8046	0.6557	1.5252	0.422
2.16	0.5525	5.2765	2.8962	1.8219	0.6464	1.5469	0.436
2.18	0.5498	5.3778	2.9238	1.8393	0.6373	1.5692	0.451
2.20	0.5471	5.4800	2.9512	1.8569	0.6281	1.5920	0.465
2.22	0.5444	5.5831	2.9784	1.8746	0.6191	1.6154	0.480
2.24	0.5418	5.6872	3.0053	1.8924	0.6100	1.6393	0.494
2.26	0.5393	5.7922	3.0319	1.9104	0.6011	1.6638	0.509
2.28	0.5368	5.8981	3.0584	1.9285	0.5921	1.6888	0.524
2.30	0.5344	6.0050	3.0845	1.9468	0.5833	1.7144	0.539
2.32	0.5321	6.1128	3.1105	1.9652	0.5745	1.7406	0.554
2.34	0.5297	6.2215	3.1362	1.9838	0.5658	1.7674	0.569
2.36	0.5275	6.3312	3.1617	2.0025	0.5572	1.7948	0.585
2.38	0.5253	6.4418	3.1869	2.0213	0.5486	1.8227	0.600
2.40	0.5231	6.5533	3.2119	2.0403	0.5401	1.8514	0.616
2.42	0.5210	6.6658	3.2367	2.0595	0.5317	1.8806	0.632
2.44	0.5189	6.7792	3.2612	2.0788	0.5234	1.9105	0.647
2.46	0.5169	6.8935	3.2855	2.0982	0.5152	1.9410	0.663
2.48	0.5149	7.0088	3.3095	2.1178	0.5071	1.9721	0.679
2.50	0.5130	7.1250	3.3333	2.1375	0.4990	2.0039	0.695
2.52	0.5111	7.2421	3.3569	2.1574	0.4911	2.0364	0.711
2.54	0.5092	7.3602	3.3803	2.1774	0.4832	2.0696	0.727
2.56	0.5074	7.4792	3.4034	2.1976	0.4754	2.1035	0.744
2.58	0.5056	7.5991	3.4263	2.2179	0.4677	2.1380	0.760
2.60	0.5039	7.7200	3.4490	2.2383	0.4601	2.1733	0.776
2.62	0.5022	7.8418	3.4714	2.2590	0.4526	2.2093	0.793
2.64	0.5005	7.9645	3.4936	2.2797	0.4452	2.2461	0.809
2.66	0.4988	8.0882	3.5156	2.3006	0.4379	2.2835	0.826
2.68	0.4972	8.2128	3.5374	2.3217	0.4307	2.3218	0.842
2.70	0.4956	8.3383	3.5590	2.3429	0.4236	2.3608	0.859
2.72	0.4941	8.4648	3.5803	2.3642	0.4166	2.4005	0.876
2.74	0.4926	8.5922	3.6015	2.3858	0.4097	2.4411	0.892
2.76	0.4911	8.7205	3.6224	2.4074	0.4028	2.4825	0.909
2.78	0.4896	8.8498	3.6431	2.4292	0.3961	2.5246	0.926
2.80	0.4882	8.9800	3.6635	2.4512	0.3895	2.5676	0.943
2.82	0.4868	9.1111	3.6838	2.4733	0.3829	2.6115	0.960
2.84	0.4854	9.2432	3.7039	2.4955	0.3765	2.6561	0.977
2.86	0.4840	9.3762	3.7238	2.5179	0.3701	2.7017	0.994
2.88	0.4827	9.5101	3.7434	2.5405	0.3639	2.7481	1.011
2.90	0.4814	9.6450	3.7629	2.5632	0.3577	2.7954	1.028
2.92	0.4801	9.7808	3.7821	2.5861	0.3517	2.8436	1.045
2.94	0.4788	9.9175	3.8012	2.6091	0.3457	2.8927	1.062
2.96	0.4776	10.0552	3.8200	2.6322	0.3398	2.9427	1.079
2.98	0.4764	10.1938	3.8387	2.6555	0.3340	2.9937	1.096

TABLE II (Suite): Ecoulement à onde de choc normale d'un gaz parfait ($\gamma = 1.4$).

M_1 $M_1 \sin\sigma$	M_2 $M_2 \sin(\sigma - \delta)$	p_2 / p_1	ρ_2 / ρ_1 V_1 / V_2 V_{n1} / V_{n2}	T_2 / T_1	p_{02} / p_{01}	A_2^* / A_1^*	$\frac{\Delta s}{R}$
3.00	0.4752	10.3333	3.8571	2.6790	0.3283	3.0456	1.114
3.02	0.4740	10.4738	3.8754	2.7026	0.3227	3.0985	1.131
3.04	0.4729	10.6152	3.8935	2.7264	0.3172	3.1523	1.148
3.06	0.4717	10.7575	3.9114	2.7503	0.3118	3.2072	1.165
3.08	0.4706	10.9008	3.9291	2.7744	0.3065	3.2630	1.183
3.10	0.4695	11.0450	3.9466	2.7986	0.3012	3.3199	1.200
3.12	0.4685	11.1901	3.9639	2.8230	0.2960	3.3778	1.217
3.14	0.4674	11.3362	3.9811	2.8475	0.2910	3.4368	1.235
3.16	0.4664	11.4832	3.9981	2.8722	0.2860	3.4969	1.252
3.18	0.4654	11.6311	4.0149	2.8970	0.2811	3.5580	1.269
3.20	0.4643	11.7800	4.0315	2.9220	0.2762	3.6202	1.287
3.22	0.4634	11.9298	4.0479	2.9471	0.2715	3.6835	1.304
3.24	0.4624	12.0805	4.0642	2.9724	0.2668	3.7480	1.321
3.26	0.4614	12.2322	4.0803	2.9979	0.2622	3.8136	1.339
3.28	0.4605	12.3848	4.0963	3.0234	0.2577	3.8803	1.356
3.30	0.4596	12.5383	4.1120	3.0492	0.2533	3.9483	1.373
3.32	0.4587	12.6928	4.1276	3.0751	0.2489	4.0174	1.391
3.34	0.4578	12.8482	4.1431	3.1011	0.2446	4.0877	1.408
3.36	0.4569	13.0045	4.1583	3.1273	0.2404	4.1593	1.425
3.38	0.4560	13.1618	4.1734	3.1537	0.2363	4.2321	1.443
3.40	0.4552	13.3200	4.1884	3.1802	0.2322	4.3061	1.460
3.42	0.4544	13.4791	4.2032	3.2069	0.2282	4.3815	1.477
3.44	0.4535	13.6392	4.2178	3.2337	0.2243	4.4581	1.495
3.46	0.4527	13.8002	4.2323	3.2607	0.2205	4.5361	1.512
3.48	0.4519	13.9621	4.2467	3.2878	0.2167	4.6154	1.529
3.50	0.4512	14.1250	4.2609	3.3150	0.2129	4.6960	1.547
3.52	0.4504	14.2888	4.2749	3.3425	0.2093	4.7780	1.564
3.54	0.4496	14.4535	4.2888	3.3701	0.2057	4.8614	1.581
3.56	0.4489	14.6192	4.3026	3.3978	0.2022	4.9461	1.599
3.58	0.4481	14.7858	4.3162	3.4257	0.1987	5.0323	1.616
3.60	0.4474	14.9533	4.3296	3.4537	0.1953	5.1200	1.633
3.62	0.4467	15.1218	4.3429	3.4819	0.1920	5.2091	1.650
3.64	0.4460	15.2912	4.3561	3.5103	0.1887	5.2997	1.668
3.66	0.4453	15.4615	4.3692	3.5388	0.1855	5.3918	1.685
3.68	0.4446	15.6328	4.3821	3.5674	0.1823	5.4854	1.702
3.70	0.4439	15.8050	4.3949	3.5962	0.1792	5.5805	1.719
3.72	0.4433	15.9781	4.4075	3.6252	0.1761	5.6773	1.736
3.74	0.4426	16.1522	4.4200	3.6543	0.1731	5.7755	1.754
3.76	0.4420	16.3272	4.4324	3.6836	0.1702	5.8754	1.771
3.78	0.4414	16.5031	4.4447	3.7130	0.1673	5.9770	1.788
3.80	0.4407	16.6800	4.4568	3.7426	0.1645	6.0801	1.805
3.82	0.4401	16.8578	4.4688	3.7723	0.1617	6.1849	1.822
3.84	0.4395	17.0365	4.4807	3.8022	0.1589	6.2914	1.839
3.86	0.4389	17.2162	4.4924	3.8323	0.1563	6.3997	1.856
3.88	0.4383	17.3968	4.5041	3.8625	0.1536	6.5096	1.873
3.90	0.4377	17.5783	4.5156	3.8928	0.1510	6.6213	1.890
3.92	0.4372	17.7608	4.5270	3.9233	0.1485	6.7348	1.907
3.94	0.4366	17.9442	4.5383	3.9540	0.1460	6.8500	1.924
3.96	0.4360	18.1285	4.5494	3.9848	0.1435	6.9671	1.941
3.98	0.4355	18.3138	4.5605	4.0157	0.1411	7.0861	1.958
4.00	0.4350	18.5000	4.5714	4.0469	0.1388	7.2069	1.975
4.02	0.4344	18.6871	4.5823	4.0781	0.1364	7.3296	1.992

TABLE II (Suite): Ecoulement à onde de choc normale d'un gaz parfait ($\gamma = 1.4$).

M_1 $M_1 \sin\sigma$	M_2 $M_2 \sin(\sigma - \delta)$	p_2 / p_1	ρ_2 / ρ_1 V_1 / V_2 V_{n1} / V_{n2}	T_2 / T_1	p_{02} / p_{01}	A_2^* / A_1^*	$\frac{\Delta s}{R}$
4.04	0.4339	18.8752	4.5930	4.1096	0.1342	7.4542	2.009
4.06	0.4334	19.0642	4.6036	4.1412	0.1319	7.5807	2.026
4.08	0.4329	19.2541	4.6141	4.1729	0.1297	7.7092	2.042
4.10	0.4324	19.4450	4.6245	4.2048	0.1276	7.8397	2.059
4.12	0.4319	19.6368	4.6348	4.2368	0.1254	7.9721	2.076
4.14	0.4314	19.8295	4.6450	4.2690	0.1234	8.1067	2.093
4.16	0.4309	20.0232	4.6550	4.3014	0.1213	8.2433	2.109
4.18	0.4304	20.2178	4.6650	4.3339	0.1193	8.3819	2.126
4.20	0.4299	20.4133	4.6749	4.3666	0.1173	8.5227	2.143
4.22	0.4295	20.6098	4.6847	4.3994	0.1154	8.6656	2.159
4.24	0.4290	20.8072	4.6944	4.4324	0.1135	8.8107	2.176
4.26	0.4286	21.0055	4.7040	4.4655	0.1116	8.9579	2.193
4.28	0.4281	21.2048	4.7135	4.4988	0.1098	9.1074	2.209
4.30	0.4277	21.4050	4.7229	4.5322	0.1080	9.2591	2.226
4.32	0.4272	21.6061	4.7322	4.5658	0.1062	9.4131	2.242
4.34	0.4268	21.8082	4.7414	4.5995	0.1045	9.5693	2.259
4.36	0.4264	22.0112	4.7505	4.6334	0.1028	9.7279	2.275
4.38	0.4260	22.2151	4.7595	4.6675	0.1011	9.8889	2.291
4.40	0.4255	22.4200	4.7685	4.7017	0.0995	10.0522	2.308
4.42	0.4251	22.6258	4.7773	4.7361	0.0979	10.2179	2.324
4.44	0.4247	22.8325	4.7861	4.7706	0.0963	10.3860	2.340
4.46	0.4243	23.0402	4.7948	4.8053	0.0947	10.5566	2.357
4.48	0.4239	23.2488	4.8034	4.8401	0.0932	10.7297	2.373
4.50	0.4236	23.4583	4.8119	4.8751	0.0917	10.9053	2.389
4.52	0.4232	23.6688	4.8203	4.9102	0.0902	11.0835	2.405
4.54	0.4228	23.8802	4.8287	4.9455	0.0888	11.2642	2.422
4.56	0.4224	24.0925	4.8369	4.9810	0.0874	11.4476	2.438
4.58	0.4220	24.3058	4.8451	5.0166	0.0860	11.6336	2.454
4.60	0.4217	24.5200	4.8532	5.0523	0.0846	11.8222	2.470
4.62	0.4213	24.7351	4.8612	5.0882	0.0832	12.0135	2.486
4.64	0.4210	24.9512	4.8692	5.1243	0.0819	12.2076	2.502
4.66	0.4206	25.1682	4.8771	5.1605	0.0806	12.4044	2.518
4.68	0.4203	25.3861	4.8849	5.1969	0.0793	12.6040	2.534
4.70	0.4199	25.6050	4.8926	5.2334	0.0781	12.8064	2.550
4.72	0.4196	25.8248	4.9002	5.2701	0.0769	13.0117	2.566
4.74	0.4192	26.0455	4.9078	5.3070	0.0756	13.2199	2.582
4.76	0.4189	26.2672	4.9153	5.3440	0.0745	13.4309	2.598
4.78	0.4186	26.4898	4.9227	5.3811	0.0733	13.6449	2.613
4.80	0.4183	26.7133	4.9301	5.4184	0.0721	13.8619	2.629
4.82	0.4179	26.9378	4.9374	5.4559	0.0710	14.0820	2.645
4.84	0.4176	27.1632	4.9446	5.4935	0.0699	14.3050	2.661
4.86	0.4173	27.3895	4.9518	5.5313	0.0688	14.5311	2.676
4.88	0.4170	27.6168	4.9589	5.5692	0.0677	14.7604	2.692
4.90	0.4167	27.8450	4.9659	5.6073	0.0667	14.9928	2.708
4.92	0.4164	28.0741	4.9728	5.6455	0.0657	15.2283	2.723
4.94	0.4161	28.3042	4.9797	5.6839	0.0647	15.4671	2.739
4.96	0.4158	28.5352	4.9865	5.7224	0.0637	15.7092	2.754
4.98	0.4155	28.7671	4.9933	5.7611	0.0627	15.9545	2.770
5.00	0.4152	29.0000	5.0000	5.8000	0.0617	16.2031	2.785
∞	0.3780	∞	6.0000	∞	0	∞	∞